Methods: GS1354-645

ALMA Thesis Planner (automated pipeline)

Overview and sequencing. The work divides into four blocks that you should run in this order, because each one fixes the scope of the next: (0) an archive and literature audit of program 2013.1.00222.T, (A) a visibility-plane photometry pipeline, (B) a quantitative injection–recovery characterization of self-calibration, and (C) an analytic derivation of the millimeter variability each jet picture predicts. Budget roughly two weeks for Step 0, eight weeks for Chapter A, eight to ten weeks for Chapter B, four to six weeks for Chapter C (which can run in parallel with B once the pipeline is stable), and the remainder for synthesis and writing. Everything runs on archival ALMA data from this one program; no new observations are involved.

Step 0 — Archive and literature audit. Query the ALMA Science Archive programmatically with ‘astroquery.alma‘ on the project code 2013.1.00222.T, retrieve the full member-OUS listing, and download the raw ASDMs together with the delivered calibration products and the QA2 reports. Build a single audit table with one row per execution block containing: target triggered, UTC date and start time, receiver band, total and on-source integration time, number of antennas and array configuration (report the resulting synthesized beam and maximum recoverable scale), spectral windows with centre frequencies and bandwidths, the identity of the bandpass, flux and phase calibrators, the phase-calibrator cycle time, the presence or absence of a check source, and the median precipitable water vapour and phase RMS from the QA2 report. For each block, make a quick continuum image from the delivered calibrated measurement set with ‘tclean‘ (‘specmode=’mfs’‘, natural weighting, ‘nterms=1‘) purely to confirm detection and record a first-pass flux density and image RMS. This table is your first deliverable and you should show it to me before proceeding.

In parallel, run the literature audit. For every source that was actually triggered, search ADS and the ATel archive for that outburst: published radio flux densities (VLA, ATCA, AMI, VLBI) within a few days of the ALMA trigger date, the X-ray spectral-state classification for that epoch, and the adopted distance and inclination. Tabulate what exists, with citations and error bars, and tabulate explicitly what does not exist. The astronomical interpretation at the end of the thesis is built only from what this table contains; if it is thin, we scale the interpretation back and say so in print.

Chapter A — the measurement pipeline.

A1. Restoration and calibration inspection. Restore each block with the delivered ‘scriptForPI.py‘ under the matching CASA version (use a containerized or separate CASA install per required version so that restorations are reproducible). Do not simply accept the calibrated product: inspect it. Use ‘plotms‘ and the ‘analysisUtils‘ package to examine the bandpass solutions per antenna and spectral window, the phase-calibrator gain phases as a function of time (this is your direct handle on atmospheric coherence), the amplitude gains, and the $T_{\{}rm sys}$ spectra. From the phase-calibrator gain phases, compute the residual phase RMS $\{}phi_{\{}rm rms}$ per antenna over the phase-referencing cycle time and convert it to an expected coherence loss $\langle A\rangle/A_0 \simeq \exp(-\phi_{\rm rms}^2/2)$; record this per block. Also record the weather (PWV time series) and the elevation track of target and calibrators. Then re-derive the visibility weights with `statwt` on line-free channels and verify them independently by differencing adjacent channels or adjacent integrations and comparing the measured scatter to the nominal weights — every error bar in the thesis descends from these weights, so validate them before using them.

A2. Visibility-plane point-source photometry. Split out the continuum spectral windows, per sideband, with `mstransform`, averaging channels only within a spectral window and never across sidebands. Then measure the flux in the uv plane, not the image plane. First determine the source position: fit a point-source model to the full-track data with `uvmodelfit` (or with your own least-squares fit using `casatools.ms` to pull `DATA`, `UVW`, `WEIGHT` and `FLAG` into numpy), solving for flux and for offsets (l,m). Then phase-rotate the visibilities to that fitted position with `fixvis`/`phaseshift`, so the source sits at the phase centre.

The critical point for low SNR: once the source is at the phase centre, the *real part* of the weighted-mean visibility is an unbiased estimator of the flux density, whereas the visibility amplitude $|V|$ is positively biased (Rice distribution) and will manufacture spurious detections in short time bins. Your flux estimator per bin is therefore $$S = \frac{\sum_i w_i\,\mathrm{Re}(V_i)}{\sum_i w_i},\qquad \sigma_S = \left(\sum_i w_i\right)^{-1/2},$$ summed over all baselines, channels and integrations in the bin, with w_i the validated weights. Carry the imaginary part through as a null test: it should be consistent with zero with the same σ_S in every bin, and you will plot it alongside every light curve. Write this as a standalone, version-controlled Python module with unit tests against simulated visibilities of known flux, so that it can be run identically on target, check source and calibrators.

A3. Noise-driven time binning. Measure the per-integration sensitivity empirically from the scatter of $\mathrm{Re}(V)$ on the target, and confirm it scales as $t^{-1/2}$ over the block. Choose the bin length Δt per block so that each bin reaches SNR ≈ 5–10 on the mean flux; if the source is bright enough, produce a small ladder of binnings (for example the shortest bin reaching SNR ≈ 5, and factors of 2, 4, 8 above it) so that variability can be probed at several timescales. State in the text, *before* quoting any variability result, the resulting time resolution and the corresponding minimum detectable fractional rms per block, computed as $\sigma_S/(S\sqrt{N_{\rm bin}})$ for the ensemble.

Chapter B — self-calibration as a measurement problem.

B1. Two parallel pipelines. Run every block through photometry twice. Path 1: standard phase-referenced calibration only, no self-calibration. Path 2: phase-only self-calibration on the target. Use `gaincal` with `calmode='p'`, `gaintype='G'`, `combine='spw'` to maximize per-solution SNR, `minsnr` around 2–3, `refant` chosen from the antenna with the most complete and stable solutions, and a model from a first `tclean` iteration on the full track. Try a ladder of solution intervals — scan-length (`solint='inf'`), then 60 s, 30 s, 10 s, and `'int'` — and for each record the number and fraction of failed/flagged solutions, the change in image RMS, and the change in fitted flux. Amplitude self-calibration is never used, because it absorbs exactly the source flux variation we are trying to measure; state this explicitly. Compare the two paths block by block: total flux, per-bin scatter, and measured fractional rms.

B2. Injection–recovery and the transfer function — the core methods result. Take the calibrated, pre-selfcal visibilities and inject a known time-dependent flux modulation by multiplying the target visibilities by $\{}left[1+m(t)\{}right]$ in a copy of the measurement set, writing back through the ‘casatools.ms‘/‘table‘ tools. Use two families of $m(t)$: (i) sinusoids with periods $\{}tau$ spanning a decade below to a decade above the self-calibration solution interval — say 5 s to 3000 s — and (ii) shot-like flares with fast rise and exponential decay of comparable characteristic timescales, which is the morphology internal shocks predict. Use injected fractional-rms amplitudes from 2% to 50%, logarithmically spaced. For each $(\{}tau, \{}text{amplitude}, \{}text{waveform})$ cell run 50–100 realizations with randomized phase and, where relevant, randomized flare arrival times, and push each one through *both* pipeline paths end to end, including re-deriving the self-calibration solutions from the modulated data (this is essential — the whole point is that selfcal solved on modulated data partly absorbs the modulation).

Define the transfer function $$T(\{}tau) = \{}frac{\{}text{recovered fractional rms}}{\{}text{injected fractional rms}}$$ measured separately for each path, and report it with its scatter across realizations, and its dependence on injected amplitude (check for the nonlinearity you should expect once the modulation is large enough to bias the selfcal model itself). This single curve does three jobs: it lets you choose the solution interval on evidence rather than folklore; it tells you empirically at which timescales selfcal's decorrelation gain exceeds its signal loss for *this* data; and it converts every rms measurement and every upper limit into a corrected or explicitly bounded quantity. Every number in Chapter A's light curves is subsequently quoted both raw and $T(\{}tau)$-corrected.

B3. Independent systematics bounds, kept as bounds. Where a check source exists, push it through the identical pipeline — it is the only genuine control, and its measured fractional rms is your systematics floor. Where no check source exists, construct two loose upper bounds. First, solve gains on the phase calibrator and apply them to the bandpass/flux calibrator treated as if it were an unknown target, then run the same photometry; the resulting apparent variability bounds the gain-transfer systematic. Second, perform odd-to-even phase-calibrator scan transfer (solve on odd scans, interpolate onto even scans), which doubles the interpolation interval and therefore *overestimates* decorrelation. State both as upper bounds and not as corrections, and say why: calibrators are 10–100 times brighter than the target and sit at different elevation and sky position, so their coherence properties are not identical. Where no check source exists, every variability statement in the thesis is written as a limit.

B4. Variability statistics. From the final light curves compute the normalized excess variance, $\{}sigma_{\{}rm XS}^2 = \{}left(S^2_{\{}rm var} - \{}overline{\{}sigma_{\{}rm err}^2}\{}right)/\{}bar{S
with uncertainties from the standard bootstrap plus the Monte Carlo realizations of B2; a first-order structure function; and, if a block is long enough, a Lomb–Scargle periodogram to check where in timescale any power sits. Convert non-detections into 3$\{}sigma$ fractional-rms upper limits *after* dividing by $T(\{}tau)$. Plot the final result as fractional rms (or limit) versus timescale, with the systematics floor from B3 drawn as a shaded band.

B5. Sideband ratio and its error budget. Measure the flux in each sideband separately with the same estimator. Before interpreting the ratio, write out the propagation explicitly: for a Band 6 continuum setup spanning roughly 224–240 GHz, $\{}ln(\{}nu_2/\{}nu_1)\{}approx 0.07$, so $\{}sigma_\{}alpha \{}approx 1.4\{},(\{}sigma_F/F)/0.07$, and a useful $\{}sigma_\{}alpha\{}approx0.3$ requires ~1.5% *relative* sideband accuracy. Estimate the relative sideband systematic empirically by measuring the sideband ratio of the check source and of the phase calibrator through the identical pipeline. Report the target sideband ratio with the systematic term shown separately and dominant. Compute a band-to-band spectral index only if Step 0 revealed a genuine second, well-separated band, in which case the limiting term becomes the absolute flux-scale accuracy, which you take from the ALMA Calibrator Source Catalogue monitoring of the flux calibrator nearest in time. Absolute flux-scale error is always quoted separately from statistical error, and is never propagated into intra-block variability, since it is common-mode.

Chapter C — deriving what each picture predicts. Do these calculations yourself in a documented notebook, anchored to the distance, published radio flux density and X-ray state you recorded in Step 0.

C1. Compact conical jet. Adopt the Blandford–Königl self-absorbed conical jet. Using the source's published flat-spectrum radio flux and distance, normalize the jet so that it reproduces that radio flux, then locate the $\tau_\nu = 1$ photosphere as a function of frequency using $z_\nu \propto \nu^{-1}$, and evaluate $z_{230\,\rm GHz}$. Propagate the parameter uncertainties (jet opening angle, magnetic-field and particle normalization, bulk Lorentz factor, inclination) through to a range on z_ν. Then compute the smearing kernel: the light-crossing time of the emitting region, $t_{\rm lc}\sim z_\nu/(c\,\delta)$ with δ the Doppler factor, and the time for a perturbation launched at the jet base to traverse the photosphere. Convert that kernel into a low-pass filter and express the result as a predicted attenuation of injected fractional rms versus timescale. Expect $z_{230\,\rm GHz}\sim10^{10}$–$10^{11}$ cm — light-seconds, not light-minutes — so the corner frequency lands at seconds to tens of seconds and the prediction is that millimeter emission tracks jet-base injection with only modest damping over the minute-to-hour range this data can probe. Show the corner timescale's sensitivity to each parameter.

C2. Internal shocks / discrete ejecta. Implement a minimal one-dimensional shell code: launch shells at a cadence set by inner-accretion-flow variability with a Lorentz-factor distribution of prescribed mean Γ and spread $\Delta\Gamma/\Gamma$, propagate them ballistically, find pairwise collision radii $z_{\rm coll}\simeq 2c\,\Delta t_{\rm launch}\,\Gamma^2$ (with the appropriate two-shell expression), and compute where each collision becomes optically thin at 230 GHz relative to the C1 photosphere. Convert collisions into flares with rise times set by the shell-crossing time and decays set by adiabatic and radiative losses, and synthesize the resulting 230 GHz light curve for a grid of $(\Gamma, \Delta\Gamma/\Gamma, \Delta t_{\rm launch})$.

C3. Comparison products. From C1 and C2, produce predicted fractional-rms-versus-timescale curves with their parameter-sensitivity envelopes, and identify the distinguishing signature — smooth red-noise wander from filtered injection versus discrete flare structure, and the timescale at which each puts its power. Overplot these on the measured curve from B4, including the $T(\tau)$-corrected detection thresholds, and state plainly which timescales the data discriminate on and which they do not.

Astronomical placement. Place the measured Band 6 flux density (or upper limit), with its absolute flux-scale error, on the published radio-to-infrared spectrum of that outburst, using only the literature values recorded in Step 0 and ordinary citation. Extrapolate the flat optically thick radio spectrum to 230 GHz and quote the mm-to-radio ratio; use the result to bound where the optically thick–thin break can lie, and propagate that bound into the C1 jet normalization. The separately awarded quasi-simultaneous radio, optical–IR, Swift/INTEGRAL/Suzaku/NuSTAR and Fermi data are not part of this dataset, and no conclusion may rest on reduced products from them.

If the source is faint or undetected. Run the identical pipeline and quote a deep, carefully bounded mm upper limit placed on the published radio-to-IR spectrum, constraining how far the flat compact-jet spectrum extends. In this branch, execute B1–B4 in full on the check source and calibrators, which are always bright enough, so that the transfer function and systematics characterization stand as the measurement result.

Tools, reproducibility and deliverables. CASA 6.x with the matching ALMA pipeline versions for restoration; ‘analysisUtils‘ for calibration diagnostics and weather; ‘astroquery.alma‘ for archive queries; ‘casatools‘/‘casatasks‘ driven from Python for the photometry, injection and Monte Carlo loops; numpy, scipy, astropy, emcee and matplotlib for fitting, statistics and plotting. Keep everything in a git repository with one driver script per figure and per table, a fixed random seed for every Monte Carlo, and a ‘make‘-style top-level script that regenerates the full thesis from the archived measurement sets. The deliverables are: the program audit table and literature-availability table; the documented, reusable \$uv\$-plane photometry module with both calibration paths; homogeneous per-block and per-sideband flux densities with statistical error, separately quoted absolute flux-scale error, and the bounded systematics floor; the measured self-calibration transfer function \$T(\\{}tau)\$; light curves with pre-stated, \$T(\\{}tau)\$-corrected detection thresholds and fractional-rms detections or limits across the accessible timescales; and the derived rms-versus-timescale predictions for the conical-jet and internal-shock pictures with the comparison against the measurement.